CLASSROOM CONTACT PROGRAMME
(Academic Session : 2025 - 2026)

ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSE, LEADER, ACHIEVER COURSE SRG MAJOR & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 02-06-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	4	1	1	2	3	2	1	1	2	3	3	3	4	3	4	2	4	3	1	3	2	3	2	2	3	2	2	2	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	1	3	2	3	1	1	1	3	3	2	3	2	1	2	2	3	2	4	1	1	4	2	2	4	4	2	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	2	2	2	4	4	2	4	3	1	3	1	4	3	2	3	2	2	3	4	2	3	3	2	1	2	3	1	3	2
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	3	1	2	2	4	2	4	2	4	2	4	4	1	3	2	1	3	4	4	2	1	1	2	2	3	3	4	2
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	2	2	3	4	3	4	1	2	1	1	1	2	2	2	2	4	2	2	4	4	1	1	2	4	2	1	4	1	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	4	1	2	2	4	1	2	4	1	3	1	2	4	2	1	3	2	3	2	2	4	3	1	1	1	4	2	3	3

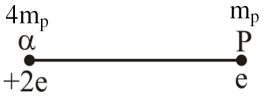
HINT – SHEET

1. Ans (1)

$$t = \frac{x}{u}, y = \frac{1}{2}\left(\frac{qE}{m}\right)\left(\frac{x}{u}\right)^2$$

2. Ans (4)

$K_i + U_i = K_f + U_f$



$$0 + \frac{K(2e)e}{r} = K_\alpha + K_P + 0 = K_\alpha + 4K_\alpha$$

$$K_\alpha = \frac{2}{5}\frac{Ke^2}{r} = \frac{2}{5} \times \frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$$

$$\boxed{K_\alpha = \frac{e^2}{10\pi\epsilon_0 r}}$$

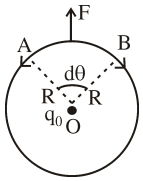
3. Ans (1)

$$E \times 4\pi x^2 = \frac{\int_0^x \frac{Q}{ar^2} \times 4\pi r^2 \times dr}{\epsilon_0}$$

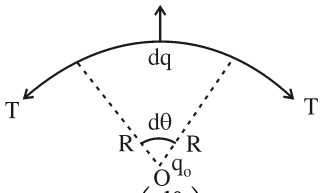
$$E = \frac{Q}{ax\epsilon_0}$$

Put $x = \frac{a}{2}$; $E = \frac{2Q}{a^2\epsilon_0}$

4. Ans (1)



Take small segment AB



$$F = 2T\sin\left(\frac{d\theta}{2}\right)$$

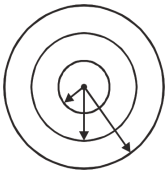
$$\frac{kq_0dq}{R^2} = 2T\left(\frac{d\theta}{2}\right) \quad [\because d\theta \text{ is very small}]$$

$$\frac{kq_0qd\theta}{2\pi R^2} = T\,d\theta$$

$$T = \frac{kq_0q}{2\pi R^2}$$

$$\left[dq = \frac{qR}{2\pi R}d\theta \qquad dq = \frac{qd\theta}{2\pi}\right]$$

5. Ans (2)



$$\frac{Q_1}{4\pi R^2} = \frac{Q_1+Q_2}{4\pi\times(4R^2)} = \frac{Q_1+Q_2+Q_3}{4\pi\times 9R^2}$$

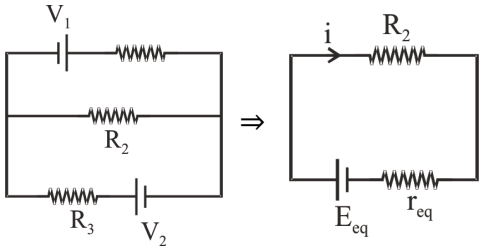
$8Q_1 = Q_2 + Q_3$

$Q_3 = 5Q_1$

$3Q_1 = Q_2$

$Q_1 : Q_2 : Q_3 = Q_1 : 3Q_1 : 5Q_1 = 1 : 3 : 5.$

6. Ans (3)



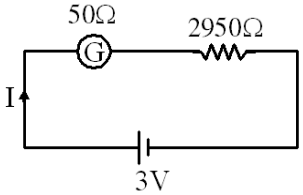
$\because$ $i = 0$

$\therefore E_{eq} = 0$

$$\frac{\frac{V_2}{R_3}-\frac{V_1}{R_1}}{\frac{1}{R_3}+\frac{1}{R_1}} = 0$$

$$\therefore \frac{V_2}{V_1} = \frac{R_3}{R_1}$$

7. Ans (2)



$$I = \frac{3}{3000} = \frac{1}{1000}\text{ A}$$

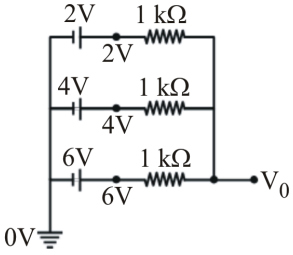
$$I' = \frac{20}{30}\times\frac{1}{1000} = \frac{3}{50+R}$$

$\Rightarrow 20\,(50 + R) = 90\,(1000)$

$\Rightarrow 100 + 2R = 9000$

$\Rightarrow R = 4450\ \Omega$

8. Ans (1)



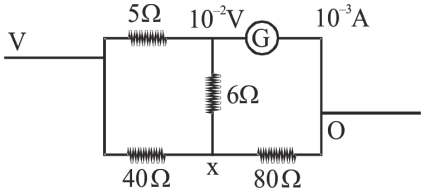
By nodal analysis

$$\frac{V_0-2}{1k\Omega}+\frac{V_0-4}{1k\Omega}+\frac{V_0-6}{1k\Omega} = 0$$

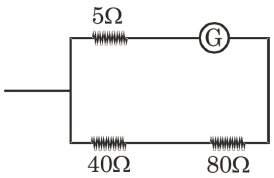
$3V0 - 12 = 0$

$V_0 = 4$

9. Ans (1)



whest stone bridge



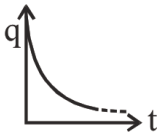
$15\times10^{-3} = 120\times i$

$$i = \frac{1}{8}\times10^{-3}\text{A} \Rightarrow i_T = \frac{9}{8}\text{mA}$$

$$4V = \frac{120\times15}{120+15}\times\frac{9}{8}\text{mA}$$

$$= \frac{40}{3}\times\frac{9}{8} = 15\text{ mV}$$

10. Ans (2)



$$C = \frac{\varepsilon_0 A}{d+vt}$$

$$q = CV = \frac{\varepsilon_0 A}{d+vt}V$$

$q(d + vt) = \varepsilon_0 AV = \text{Const}$

11. Ans (3)

Intial energy of capacitor

$U_i = \frac{Q^2}{2C}$

$= \frac{1}{2} \times \frac{120P \times 120P}{12P} = 600\ pJ$

Since battery is disconnected so charge remain same.

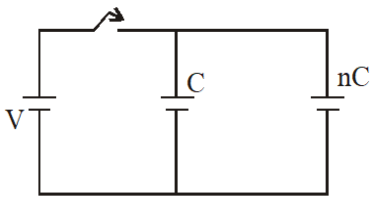
Final energy of capacitor

$U_f = \frac{1}{2}\frac{Q^2}{C} = \frac{U_i}{K} \approx 92\ pJ$

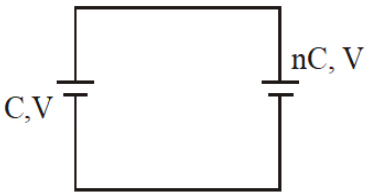
$W + U_f = U_i$

$W = 508\ pJ$

12. Ans (3)



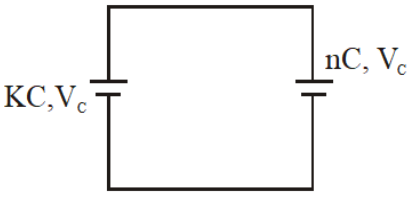
After fully charging, battery is disconnected



Total charge of the system = CV + nCV

= (n + 1)CV

After the insertion of dielectric of constant K



New potential (common)

$V_C = \frac{\text{total charge}}{\text{total capacitance}}$

$= \frac{(n+1)CV}{KC + nC} = \frac{(n+1)V}{K + n}$

13. Ans (3)

$\vec{F} = q\left[\vec{E} + (\vec{v} \times \vec{B})\right]$

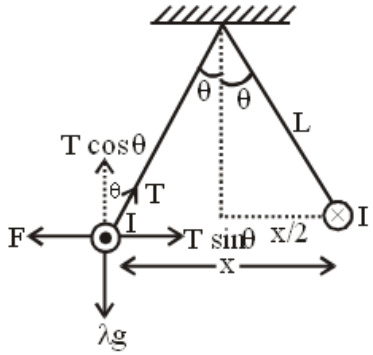
$(\vec{v} \times \vec{B}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 1 \\ 1 & 1 & -3 \end{vmatrix} = -13\hat{i} + 10\hat{j} - \hat{k}$

$\vec{F} = q\left[3\hat{i} + \hat{j} + 2\hat{k} - 13\hat{i} + 10\hat{j} - \hat{k}\right]$

$= q\left[-10\hat{i} + 11\hat{j} + \hat{k}\right]$

Hence $F_y = 11q$

14. Ans (4)



consider balance of unit length

$T\cos\theta = \lambda g$ .....(i)

$T\sin\theta = F$ .....(ii)

$\frac{\sin\theta}{\sin\theta} = \frac{F}{\lambda g}, \frac{x}{2} = L\sin\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{\mu_0 I^2}{2\pi(2L\sin\theta)}$

15. Ans (3)

%increase=

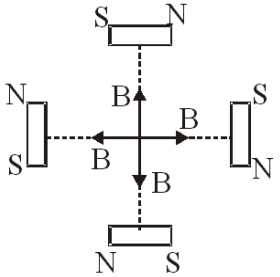
$\frac{\Delta B}{B_0} = \frac{B_m - B_0}{B_0} = \frac{\mu_r B_0 - B_0}{B_0} \times 100\%$

$= (\mu_r - 1) \times 100\% = (\chi + 1 - 1) \times 100\%$

$= 6.8 \times 10^{-5} \times 100 = 6.8 \times 10^{-3}$

$= 0.0068\%$

16. Ans (4)



17. Ans (2)

$\vec{F} = q\left(\vec{V} \times \vec{B}\right)$

$\therefore \vec{F}$ is perpendicular to $\vec{V}$

$\therefore$ Work done by MF = 0

$\therefore \Delta K = 0$

KE = constant.

Also

$\vec{F} \perp \vec{B}$

But work done by MF is zero due to $\vec{F} \perp \vec{V}$.

18. Ans (4)

$P = Vi$

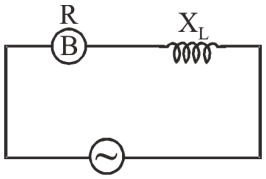
$50 = 50\ i$

$i = 1\ A$

$P = \dfrac{V^2}{R}$

$50 = \dfrac{50 \times 50}{R}$

$R = 50\Omega$



$(200) = \sqrt{(50)^2 + X_L^2}$

$(200)^2 - (50)^2 = X_L^2$

$X_L = 194\Omega = 2\pi f L$

$L = \dfrac{194}{2 \times \pi \times 50} = 0.617\ H$

19. Ans (3)

$\Delta\omega = \dfrac{R}{L}$

$Q = \dfrac{\omega_0}{\Delta\omega} = \omega_0 \dfrac{L}{R},\ \omega_0 = \dfrac{1}{\sqrt{LC}}$

$\omega_0 = \dfrac{1}{\sqrt{3 \times 27 \times 10^{-6}}} = \dfrac{1}{9 \times 10^{-3}}$

$\dfrac{Q}{\Delta_\omega} = \dfrac{\omega_0 \frac{L}{R}}{\frac{R}{L}} = \omega_0 \dfrac{L^2}{R^2} = \sqrt{\dfrac{1}{LC}} \dfrac{L^2}{R^2}$

$= \dfrac{1}{9 \times 10^{-3}} \times \dfrac{9}{100} = 10\ s$

20. Ans (1)

$\dfrac{\varepsilon_1}{\varepsilon_2} = \dfrac{N_1}{N_2} = \dfrac{100}{10} \Rightarrow \varepsilon_2 = 22\ V$

$V_0 = \dfrac{7}{7 + 15} \times 22$

$V_0 = 7$ volt

21. Ans (3)

$i = \dfrac{Bv\ell}{R},\ P = i^2R$

22. Ans (2)

$\omega_r = \dfrac{1}{\sqrt{LC}} = \dfrac{1}{\sqrt{50 \times 10^{-3} \times 5 \times 10^{-6}}}$

$\omega = \dfrac{\omega_r}{2} = 1000$

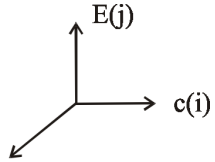
$X_L = \omega L = 1000 \times 50 \times 10^{-3} = 50\Omega$

$X_C = \dfrac{1}{\omega C} = \dfrac{1}{1000 \times 5 \times 10^{-6}} = 200\Omega$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$P = \left(\dfrac{V}{Z}\right)^2 R$

23. Ans (3)



$\therefore \vec{B}(\hat{k})$

$\Rightarrow \vec{B} = B_0 \cos(wt - kx)\hat{k}$

Now put t = 0

24. Ans (2)

$\dfrac{1}{2} c\in_0 E^2 = \dfrac{100}{4\pi \times 3^2}$

$\dfrac{1}{2} c\in_0 \left(\sqrt{\dfrac{x}{5}}E\right)^2 = \dfrac{60}{4\pi \times 3^2}$

$\Rightarrow \dfrac{x}{5} = \dfrac{3}{5} \Rightarrow x = 3$

25. Ans (2)

(a) uv rays – used for water purification

(b) x-rays used for diagnosing bone fracture

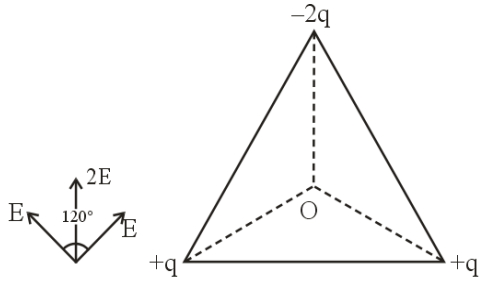
(c) Microwaves are used for mobile and radar communication

(d) Infrared waves show less scattering therefore used in foggy days

(a – ii), (b – i), (c – iii), (d – iv)

26. Ans (3)

(1) $V_{centre} = \frac{K}{r}(-2q + q + q) = 0$

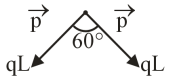


(2) for electric field -

Let $E = \frac{Kq}{r^2}$, $E_{centre} = 2E + E = 3E \neq 0$

(3) Dipole moment

$p_{net} = \sqrt{3}p \quad (\because \theta = 60°)$



27. Ans (2)

$I_1 = I_2$

$\frac{n_1 hc}{\lambda_1 At} = \frac{n_2 hc}{\lambda_2 At}$

$\frac{n_1}{n_2} = \frac{\lambda_1}{\lambda_2} = \frac{4}{3}$

Since number of photon decreases, emitted electron will be decreases and wavelength is decreasing therefore K.E. of emitted electron will increase.

28. Ans (2)

$K.E. = eV_0 = \frac{hc}{\lambda} - \phi$

$V_0 = \frac{1}{e}\left[\frac{hc}{\lambda} - \phi\right]$

29. Ans (2)

For perfectly reflecting mirror, the force exerted by the light of power P is

$F = \frac{2(power)}{c}$

For equilibirum

$F = mg = \frac{2(power)}{c}$

$Power = \frac{mg \times c}{2} = 30 \times 10^6$ W

As only 30% of the power is given to the mirror

So, $P' = 30 \times \frac{100}{30} = 100$ MW

30. Ans (4)

$Q = (m_A - m_B - m_D) \times 931.5$ MeV

$= (238.05079 - 234.04363 - 4.00260) \times 931.5$

$\Rightarrow 4.25$ Mev

31. Ans (3)

$P = \left(\frac{N}{t}\right) 250\,MeV$

$\left(\frac{N}{t}\right) = \frac{100 \times 10^6}{250 \times 1.6 \times 10^{-13}}$

$= 2.5 \times 10^{18}$

$\therefore$ Number of neutron per second

$= 2.5 \times 2.5 \times 10^{18}$

$= 6.25 \times 10^{18}$

32. Ans (2)

$\lambda_{DB} = \frac{h}{p} = \frac{h}{\sqrt{2mk}}$

$\Rightarrow \lambda_{DB} \propto \frac{1}{\sqrt{m}}$

$\Rightarrow \lambda_\alpha < \lambda_p < \lambda_e$

33. Ans (1)

$F_C = \frac{mv^2}{r}$

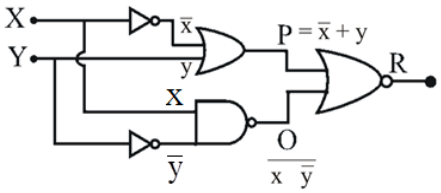
$\frac{Kq_1q_2}{r^2} = \frac{mv^2}{r}$

$mv^2r^2 = Kq_1q_2r$

$\frac{L^2}{m} = Kq_1q_2r$

$L \propto \sqrt{r}$

34. Ans (3)



$R = \overline{(\bar{x} + y) + \bar{x}\bar{y}} = \overline{(\bar{x} + y) + (\bar{x} + \bar{\bar{y}})}$

$R = \overline{\bar{x} + y} = \bar{\bar{x}}.\bar{y} = x.\bar{y}$

For R = 1 ; x = 1, y = 0

35. Ans (2)

Solution :

The potential difference across zener is 3 volt.

$\therefore$ Current through 100Ω is $= \frac{3}{100} = 0.03A$

The potential difference across 20Ω is 2 volt

$\therefore$ Current through 20Ω is $= \frac{2}{20} = 0.1A$

Current through zener is $= 0.1 - 0.03 = 0.07$

and hence the power of zener $= V_Z \times I_Z = 3 \times .07 = 0.21$ W

36. Ans (3)

From the I-V graph, the forward (knee) voltage of the diode is about

$V_D \approx 0.4$ V

The circuit current is given by:

$$I = \frac{V - V_D}{R}$$

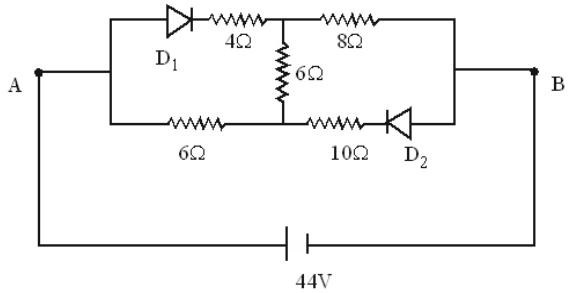
(where, V = 10V, R = 12×10^3 Ω)

$$I = \frac{10 - 0.4}{12 \times 10^3}$$

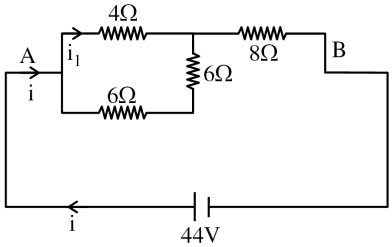
$$I = \frac{9.6}{12 \times 10^3}$$

$I = 0.8 \times 10^{-3}$ A = 0.8 mA

37. Ans (1)



We can see that diode D_1 is forward biased and diode D_2 is reverse biased, so the brunch containing 10Ω resistor becomes inactive, so diode D_2 behaves like open circuit. Now, new circuit becomes like this



$R_{AB} = [4 \parallel (6 + 6)] + 8$

$R_{AB} = (4 \parallel 12) + 8$

$$R_{AB} = \frac{4 \times 12}{8 + 12} + 8 = 11\Omega$$

$R_{AB} = 11\Omega$

$$i = \frac{V}{R_{AB}} \Rightarrow \frac{44}{11} = 4A$$

$\Rightarrow \boxed{i = 4A}$

Now, $i_1 = \left(\frac{12}{4 + 12}\right) \times 4 \Rightarrow \boxed{i_1 = 3A}$

38. Ans (1)

Angular magnification of eye piece (m_e)

$$m_e = \left(\frac{25}{5}\right) = 5 \quad \{\text{for normal adjustment}\}$$

and magnification (m) = $m_0 \times m_e$

$$\Rightarrow m_0 = \frac{m}{m_e} = \frac{-30}{5} = -6$$

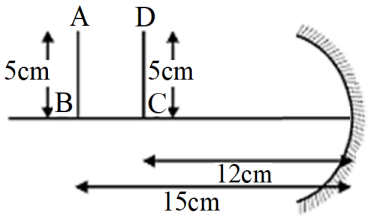
Also, $m_0 = \frac{f_0 - v_0}{f_0}$

$$\Rightarrow -6 = \frac{1.25 - v_0}{1.25}$$

$\Rightarrow v_0 = 8.75$ cm

Hence, $L = v_0 + f_e = 8.75 + 5 = 13.75$ cm

39. Ans (1)



For AB

$$\frac{1}{-15} + \frac{1}{V_1} = -\frac{1}{10} \Rightarrow V_1 = -30\text{cm}$$

For CD

$$\frac{1}{-12} + \frac{1}{V_2} = -\frac{1}{10} \Rightarrow V_2 = -60\text{cm}$$

Length of Image wire

$$= AB\left|\frac{V_1}{U_1}\right| + CD\left|\frac{V_2}{U_2}\right| + |V_2 - V_1|$$

$$= 5 \times \frac{30}{15} + 5 \times \frac{60}{12} + (60 - 30)$$

= 10 + 25 + 30 = 65 cm

40. Ans (3)

$$r = \frac{h}{\sqrt{\mu^2 - 1}}$$

The diameter D is twice the radius ($D = 2r$).

Substituting the given values $h = 1$ m and $\mu = \frac{4}{3}$:

$$D = \frac{2(1)}{\sqrt{(4/3)^2 - 1}}$$

$$D = \frac{2}{\sqrt{16/9 - 1}}$$

$$D = \frac{2}{\sqrt{7/9}} = \frac{2}{\sqrt{7}/3} = \frac{6}{\sqrt{7}}$$

41. Ans (3)

$u_1 = -5$cm, $f_1 = 5$ cm

$1/v_1 = 1/5 - 1/5 = 0$(zero) $\longrightarrow v_1 = \infty$

$u_2 = -\infty$ (distance from concave lens).

For final image to be formed on object:

$1/f_2 = 1/v_2 - 1/u_2$

$v_2 = f_2 = -10$ cm

Now, $d = |v_2| - |u_1| = 10-5$

$d = 5$ cm

42. Ans (2)

For normal adjustment

$$m = -\frac{f_0}{f_e}$$

When final image is at least distance of distinct vision from eyepiece,

$$m' = -\frac{f_0}{f_e}\left(1 + \frac{f_e}{d}\right) = 10\left(1 + \frac{5}{25}\right) = 12$$

43. Ans (3)

Intensity of incident unpolarised light = I_0

Intensity transmitted by first polaroid = $I_1 = \frac{I_0}{2}$

Intensity transmitted by second polaroid

$$I_2 = I_1 \cos^2(90° - 60°) = \frac{I_0}{2}\left(\frac{\sqrt{3}}{2}\right)^2$$

$$= \frac{3}{8}I_0 = 0.375\, I_0$$

Transmitted percentage = $\frac{I_2}{I_0} \times 100 = 0.375 \times 100$

$= 37.5\%$

44. Ans (2)

In case of diffraction at single slit, the position of minima is given by

$d \sin\theta = n\lambda$ …(1)

and for small θ : $\sin\theta = \theta = (y/D)$ …(2)

So, from Eqs. (1) and (2), we have :

$$d\left(\frac{y}{D}\right) = n\lambda, \quad \text{i.e.} \quad y = \frac{D}{d}(n\lambda)$$

So that, $y_3 - y_1 = \frac{D}{d}(3\lambda - \lambda) = \frac{D}{d}(2\lambda)$

and hence, $d = \frac{0.50 \times (2 \times 6 \times 10^{-7})}{3 \times 10^{-3}} = 2 \times 10^{-4}$m

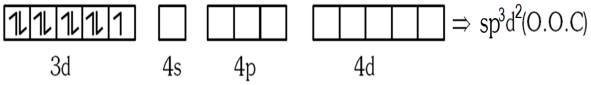
$= 0.2$ mm

45. Ans (1)

Optical Path = $\mu \times$ geometrical Path

46. Ans (2)

$Cu(H_2O)_6]^{2+} \Rightarrow Cu^{+2}(d^9)$ CN = 6 (octahedral)



$\Rightarrow sp^3d^2$(O.O.C)

n = 1 $\longrightarrow$ Paramagnetic

47. Ans (2)

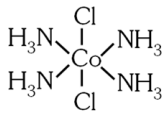
(I) $t_{2g}^3 eg^2 \Rightarrow d^5$(WFL) $\Rightarrow$ O.OC

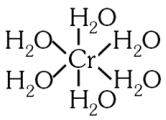
(II) $t_{2g}^6 eg^1 \Rightarrow d^7$(SFL) $\Rightarrow$ I.O.C

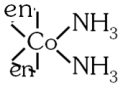
(III) $t_{2g}^3 e_g^0 \Rightarrow d^3$(Any) $\Rightarrow$ I.OC

(IV) $t_{2g}^4 e_g^2 \Rightarrow d^6$(WFL) $\Rightarrow$ O.O.C

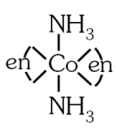
48. Ans (3)

(1) 
POS

(2) 
POS

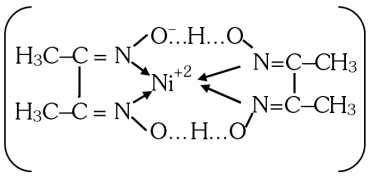
(3) 
$\Rightarrow$ Optically active

POS $[m(AA)_2a_2]$

(4) 
POS $\Rightarrow$ Optically inactive

49. Ans (2)

$[(Ni(dmg)_2] \Rightarrow Ni^{+2}\ (d^8)$



$\Rightarrow$ Extra stabilised due to intramolecular H-bond

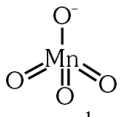
EAN = 28 − (+2) + (2×4)= 34

but still because it is highly stable complex, it will not act as an O.A

50. Ans (4)

Brown Ring complex $\Rightarrow [Fe(H_2O)_5NO]SO_4$; Here NO^+ and Fe^+ are present

51. Ans (1)



$\Rightarrow$ It has tetrahedral geometry and $109°28^1$ angle and all bonds are equal due to resonance.

52. Ans (1)

NCERT line (due to ↑ in z_{eff}, IP↑)

53. Ans (4)

(1) Stability of + 4 O.S. decreases down the group in G-14, So OA↑)

(2) Ionic character $\propto \frac{1}{\text{Polarisation}} \propto$ size of cation (Anion is same in all)

(3) Ionic character $\propto \frac{1}{\text{size of Anion}} \propto$; At no of Halogen ↑ size of anion ↑

(4) Stable $T\ell I_3$ can exist as $T\ell^{\oplus}, I_3^{\ominus}$.

54. Ans (2)

(1) Max^m CN of Si is 6

(2) BDE of $C\ell_2 > F_2$

(3) Stab : $Mn^{+2}_{(aq)} > Mn^{+3}_{(aq)}$

(4) Elements of G−15 can show many O.S.

55. Ans (2)

BP : $H_2O > H_2Te > H_2Se > H_2S$: H_2O is liq. at Room temperature.

56. Ans (4)

Question Explanation: Select the correct statements regarding Group 16 elements.

Concept: Properties and oxides of the oxygen family.

Solution:

Statement A is correct as Group 16 elements form EO_2 and EO_3 oxides which are generally acidic (Te and Po oxides show amphoteric character but are included in the general trend).

Statement B is correct because SO_2 is a reducing agent (S goes from +4 to +6) while TeO_2 is an oxidizing agent due to the inert pair effect.

Statement C is false; reducing property of hydrides increases down the group (H_2S to H_2Te).

Statement D is false; ozone (O_3) has six lone pairs of electrons.

Oxidation state change: S (+4 ⟶ +6) is oxidation;

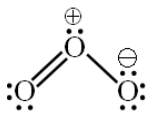
Te (+4 ⟶ 0/+2) is reduction.

2. Ozone lone pairs: Central O (1) + Terminal double bonded O (2) + Terminal single bonded O (3) = 6.

(A) All group 16 elements form oxides of the EO_2 and EO_3 type where E = S, Se, Te or PO.

(B) SO_2 is reducing while TeO_2 is an oxidising agent.

(C) The reducing property increases from H_2S to H_2Te down the group.

(D)  have six lone pairs

Final Answer: Option (4)

57. Ans (4)

AgI is insoluble in NH_4OH due to high covalent character.

58. Ans (2)

Explaining the Question :-

Question is asking about matching of compounds

Concept:-

Colour of precipitate, Colour of solution, general reaction

Solution :-

$ZnCl_2 + H_2S$ (basic) $\longrightarrow ZnS(s) + 2HCl$

ZnS(zinc sulphide) is a **white precipitate** that forms in the reaction.

$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{+2}(aq) + 2OH^-(aq)$

The **blue precpitate** of $Cu(OH)_2$ **dissolves** and the solution turns **deep blue** due to the **$[Cu(NH_3)_4]^{2+}$** complex.

$FeCl_3 + 3KSCN \longrightarrow Fe(SCN)_3 + 3KCl$

The **ferric thiocyanate complex** is **blood-red** in color.

$Pb(NO_3)_2 + 2KI \longrightarrow PbI_2 + 2KNO_3$

Lead(II) iodide (PbI_2) is the **yellow precipitate**.

Final Answer :- 2

59. Ans (2)

Forms soluble complex ion $[PbCl_4]^{2-}$.

60. Ans (4)

Chromite are $\Rightarrow FeO.Cr_2O_3$ is used to prepare $K_2Cr_2O_7$ and all are correct option.

61. Ans (1)

NCERT line.

62. Ans (2)

Explain Question:

Order (n) is found using how rate change in with change in pH for the rate law $\left(\frac{dx}{dt}\right) = k[H^+]^n$.

Concept:

$pH = -\log[H^+]$, rate $\propto [H^+]^n$

Solution:

Given : rate becomes 100 times when pH changes from 2 to 1.

at pH = 2: $[H^+]^1 = 10^{-2}$ M

at pH = 1: $[H^+]^2 = 10^{-1}$ M

$Rate_1 = K(10^{-2})^n$

$Rate_2 = K(10^{-1})^n$

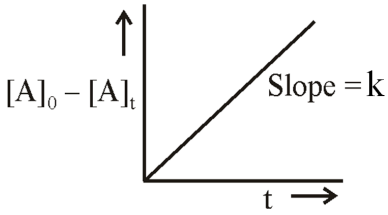
given $rate_2 = 100 \times Rate$

so: $100 = \left(\frac{10^{-1}}{10^{-2}}\right)^n = (10^1)^n$

$100 = 10^n = 2$

Final answer: option (2)

63. Ans (2)



$[A]_0 = [A]_t + kt$

$= 0.05 + 0.02 \times 30$

$= 0.65 \text{ mol } L^{-1}$

64. Ans (2)

$\frac{1}{t}\log\frac{1}{1/3} = \frac{1}{0.5t}\log\frac{1}{A_t}$

65. Ans (4)

Question Explanation: We need to derive the expression for rate constant k for the first-order gaseous reaction A ⟶ 3B. where the initial pressure is P_0 and total pressure after time t is P_r.

Concept: This question is based on first-order kinetics and pressure-extent of reaction relationship.

Solution:

Reaction:

$A \longrightarrow 3B$

Species	Initial Pressure	Change	Final Pressure
A	P_0	$-P_0\alpha$	$P_0(1-\alpha)$
B	0	$+3P_0\alpha$	$3P_0\alpha$
Total (P_r)	-	-	$P_T = P_0(1+2\alpha)$

From total pressure relation: $\alpha = \dfrac{P_T/P_0 - 1}{2}$

For a first-order reaction: $kt = \text{In}\left(\dfrac{P_0}{P_A}\right)$

Substituting $P_A = P_0(1-\alpha) = \dfrac{3P_0 - P_r}{2}$:

$k = \dfrac{1}{t}\text{ In}\left(\dfrac{P_0}{P_A}\right) = \dfrac{1}{t}\text{ In}\left(\dfrac{2P_0}{3P_0 - P_T}\right)$

Final answer: option (4)

66. Ans (4)

$w = \dfrac{E}{96500} \times i \times t$

$= \dfrac{98}{96500} \times 1.25 \times 1930 \times 60$

$= 147\text{ g}$

67. Ans (2)

Given electron potential data is

$Cl_2 + 2H_2O \longrightarrow 2ClO^- + 4H^+ + 2e^-$;

$E° = -1.61$ volt

$ClO^- + 2H_2O \longrightarrow ClO_3^- + 4H^+ + 4e^-$;

$E° = -0.50$ volt

For reaction (ii) and (iv) E° is positive. Thus these are spontaneous.

68. Ans (4)

Equivalent of deposit Cu^{+2} =

$\dfrac{It}{F} = \dfrac{4.6 \times 30 \times 60}{96500} = 0.085$

Moles of Cu^{+2} deposit = $\dfrac{0.085}{2} = 0.0425$

Molarity of Cu^{+2} deposit = $\dfrac{0.0425}{0.5} = 0.085$

Remain Molarity = 0.6 − 0.085 = 0.51

69. Ans (3)

$H_2 \longrightarrow 2H^+$

10^{-5} M

$2H^+ \longrightarrow H_2$

10^{-3} M

$2H^+ \longrightarrow 2H^+$

$10^{-3}\text{M} \quad 10^{-5}\text{M}$

$E = E^0 - \dfrac{0.059}{2}\log\left(\dfrac{10^{-5}}{10^{-3}}\right)^2 = 0.118$ volt

70. Ans (1)

$\Delta T = i \times K_b \times m$

$i \times m$ of Na_2SO_4 is highest, hence its boiling point will also be highest.

71. Ans (3)

$\Delta T_f = \dfrac{1000\,K_f\,w}{m\,W}$

$\Rightarrow \quad 9.3 = \dfrac{1000 \times 1.86 \times 50}{62 \times W}$

$\Rightarrow W = 161.29\text{ g}$

$\therefore$ Separated ice = 200 −161.29 = 38.71g

72. Ans (1)

$\Delta T_b = K_b \times m$

$.01 = 0.5 \times n \times 10$

$\text{moles} = \frac{1}{500}$

$\text{molecules} = \frac{1}{500} \times 6.023 \times 10^{23}$

$= 1.2 \times 10^{21}$ molecules

73. Ans (4)

$\pi \propto M \quad \pi_{final} = \frac{\pi_1 v_1 + \pi_2 v_2}{v_1 + v_2} = \frac{1.64v + 2.46v}{2v}$

$\pi_{final} = 2.05$ atm

74. Ans (3)

We know that

$Y_A = \frac{P_A^o X_A}{P_{total}}$ or $\frac{Y_A}{X_A} = \frac{P_A^o}{P_{total}}$

$\because P_A^o > P_{total}$ so $\frac{Y_A}{X_A} > 1$ or $Y_A > X_A$

75. Ans (2)

Explain Question:

We are asked: Which statement about colloidal solutions are correct?

Concept:

- Peptization: conversion of precipitate into colloidal sol.
- Lyophilic sols: mostly organic, highly solvated, reversible.
- Lyophobic sols: mostly inorganic, poorly solvated, irreversible, need special preparation methods.

Solution:

(i) Correct: Peptization is the conversion of a fresh precipitate into a colloid.

(ii) Incorrect: Lyophilic sols are typically organic (starch, gum). Inorganic sols are usually lyophobic.

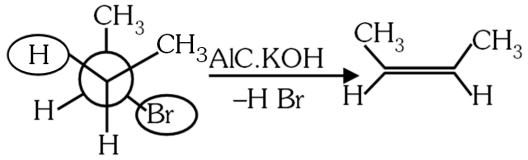
(iii) Correct: Lyophilic means "solvent-loving," so particles are highly solvated.

(iv) Incorrect: Lyophobic sols are irreversible. Lyophilic sols are reversible.

Thus, statements (i) and (iii) are correct.

Final Answer: Option (2)

76. Ans (3)



77. Ans (2)

1° halide with best L.G. favour S_N2

78. Ans (2)

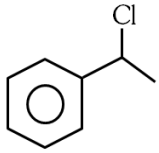
$S_N1 \propto$ stability of $C^{\oplus}$

79. Ans (3)

have chiral carbon.

80. Ans (4)

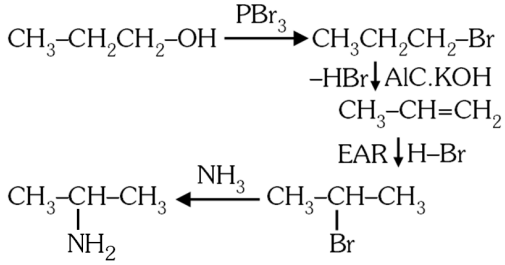
Benzyl halide give S_N1, S_N2 in good yield.



Also give Elimination rx^n.

Aryl halide show S_NAr rx^n.

81. Ans (2)



82. Ans (3)

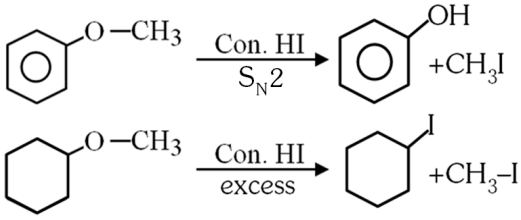
williumson ether synthesis is an example of

S_N2 –rx^n.

83. Ans (3)

Fact Based

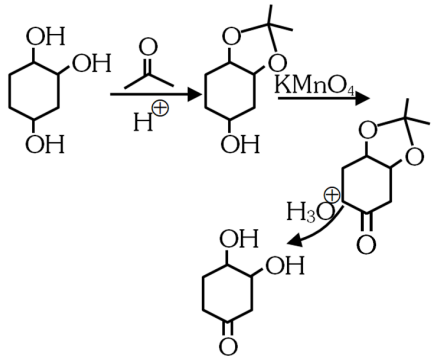
84. Ans (2)



85. Ans (1)

These reagent reduces Aldehyde into Alcohol.

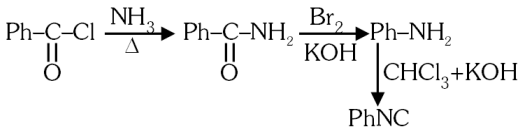
86. Ans (2)



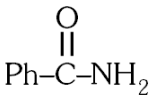
87. Ans (3)

Both give Iodoform test

88. Ans (1)



89. Ans (3)

 give Hoffman bromamide reaction & reacts with CH_3MgBr and give CH_4 gas.

90. Ans (2)

to remove S^{-2} & $\overset{\ominus}{C}N$

91. Ans (1)

NCERT-XII Pg # 201

92. Ans (3)

NCERT-XII Pg. # 199

93. Ans (3)

NCERT XII Pg. # 195

94. Ans (1)

NCERT-XII Pg.# 197

95. Ans (2)

NCERT-XII, Pg. # 192

96. Ans (2)

NCERT XII, Page # 210, 213

97. Ans (4)

NCERT XII, Page # 209

98. Ans (2)

NCERT-XII Pg. # 207, 208, 210, 213

99. Ans (4)

NCERT-XII Pg. # 210, 211

100. Ans (2)

NCERT-XII Pg. # 207

101. Ans (4)

NCERT-XII Pg. # 224, 225

102. Ans (2)

Question Explanation :

Identify which of the given options is not a type of weed.

Concept :

This question is based on Alien species invasions.

Solution :

Nile perch is not example of weed.

Hence, Option (2) is correct

103. Ans (4)

NCERT (XII) ; Page No. # 221, 222

104. Ans (4)

Question Explanation: Calculate log S from a given equation.

Concept: Species-area relationship

Solution: Using the equation, logS = logC + ZlogA, substitute the given values: logS = 0.8 + 0.3(4) = 0.8 + 1.2 = 2.

Final Answer: 2

105. Ans (1)

NCERT-XII Pg. # 218, 220, 223, 224, 225

106. Ans (3)

NCERT Pg. # 59

107. Ans (2)

NCERT (XI) Page No. 64

108. Ans (1)

NCERT (XI) Page No. 62, 67, 68

109. Ans (3)
NCERT Pg. No. # 68, 69

110. Ans (4)
NCERT-XI Pg. # 76

111. Ans (4)
NCERT-XI, Pg. # 76

112. Ans (2)
NCERT XI Pg. No.#77.

113. Ans (1)
NCERT (XI) ; Page No. # 72, 73

114. Ans (1)
NCERT Pg. # 65, 66

115. Ans (2)
NCERT-XI, Pg. # 11

Hint:

Wind pollination + self-incompatibility maintains genetic diversity.

NCERT Page No.# 13

116. Ans (2)
NCERT Pg. # 18, 19

117. Ans (3)
NCERT Pg. # 19

118. Ans (3)
NCERT-XII, Pg. No. # 16

119. Ans (4)
NCERT-XII, Pg. # 8

120. Ans (2)
NCERT, Pg. # 20

121. Ans (1)
NCERT Pg # 53

122. Ans (2)
NCERT, Pg. # 72

123. Ans (2)
NCERT-XII, Pg. # 53, 56, 57 (E)

NCERT-XII, Pg. # 60, 62, 63 (H)

124. Ans (3)
NCERT page 85

125. Ans (4)
Solution : Both DNA and RNA are able to mutate, but DNA mutates at a much faster rate.

Final Answer : (4)

126. Ans (3)
Salient features of human genome.

127. Ans (4)
NCERT-XII, Pg. # 93, 94

128. Ans (1)
NCERT Pg. # 64

129. Ans (2)
NCERT XII, Pg. # 74
Mother should be carrier

130. Ans (1)
NCERT XII, Pg. # 61
AB^{+ve} has 3 Antigens A, B and D.

131. Ans (1)
NCERT XII, Pg. # 97
some mutations can be useful.

132. Ans (1)
NCERT XII, Pg. # 104
Expressed sequence tags

133. Ans (2)
NCERT XII, Pg. # 83
2 molecules each of H2A, H2B, H3 and H4

134. Ans (2)
NCERT XII, Pg. # 97
RNA has additional OH and uracil.

135. Ans (2)
NCERT XII, Pg. # 60
True breeding shows expression for several generations.

136. Ans (2)
NCERT-XII Pg. # 168,170

137. Ans (4)
NCERT-XII Pg. # 170

138. Ans (2)
NCERT Page no. 168, 172, 173, 174

139. Ans (2)
NCERT Pg. # 163, 164

140. Ans (4)
NCERT Page No. 169

141. Ans (4)
NCERT Pg. # 183, 184, 185

142. Ans (1)
NCERT-XII, Pg. # 179

143. Ans (1)
NCERT-XII, Pg. # 180

144. Ans (2)
NCERT Pg. # 199

145. Ans (4)
NCERT-XII, Pg. # 157

146. Ans (2)
NCERT-XII, Pg. # 151, 153

147. Ans (1)
NCERT, Pg.# 156

148. Ans (4)
NCERT Pg. No. # 33, Fig. 2.8

149. Ans (1)
NCERT Pg. No. # 30

150. Ans (1)
NCERT Pg. No. # 35-38

151. Ans (4)
NCERT Pg. No. # 33

152. Ans (4)
NCERT-XII, Pg. # 43

153. Ans (1)
NCERT Pg. # 39

154. Ans (2)
NCERT Pg.# 31

155. Ans (2)
NCERT Pg.# 39

156. Ans (4)
NCERT Pg. # 42

157. Ans (1)
NCERT Pg.# 29

158. Ans (2)
NCERT Pg. # 26

159. Ans (4)
NCERT (XII) ; Page No. # 35

160. Ans (1)
NCERT Pg. No. # 42, 46

161. Ans (3)
NCERT Pg. No. # 48

162. Ans (1)
NCERT, Pg. # 43, 44, 45

163. Ans (2)
NCERT Pg.# 46

164. Ans (4)
NCERT Pg. # 43

165. Ans (2)
NCERT Pg.# 142-143

166. Ans (1)
NCERT Pg.# 141-142

167. Ans (3)
NCERT, Pg. # 131-133

168. Ans (2)
NCERT, Pg. # 135-141

169. Ans (3)
NCERT Pg. # 132

170. Ans (2)
NCERT Page: 136

171. Ans (2)
NCERT Page: 135

172. Ans (4)
NCERT-XII, Pg. No. # 141

173. Ans (3)

NCERT, Pg. # 135

174. Ans (1)

NCERT Pg. # 121

175. Ans (1)

NCERT Pg. # 121/123/124

176. Ans (1)

NCERT Pg. # 118-119

Natural selection favours organism with advantageous traits.

177. Ans (4)

NCERT Pg.# 124

178. Ans (2)

NCERT Pg.# 111

179. Ans (3)

NCERT Pg. # 124

180. Ans (3)

Molecular evidence